Homework 3

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1. Prove or find a counterexample: Let (X, d) be a metric space. Then for any $x \in X$ and $r > 0$, the set

$$\{y \in X : d(x, y) \leq r\}$$

is closed.

Proof. Fix $x \in X$, and define $A = \{y \in X : d(x, y) \leq r\}$. If A has no limit points, then it is vacuously closed. Suppose, then, that y is a limit point of A. Then given any $n \in \mathbb{Z}^+$, $B_{1/n}(y) \cap A \setminus \{y\} \neq \varnothing$. Let $z \in B_{1/n}(y) \cap A \setminus \{y\}$. Then by definition $d(x, z) \leq r$, so we have

$$d(x, y) \leq d(x, z) + d(y, z) < r + \frac{1}{n}.$$

We can then take the infimum over all n to get that $d(x, y) \leq r$, and thus $y \in A$ as desired. $\square$

2. Show that if x is a limit point of $A \cup B$ if and only if x is either a limit point of A or a limit point of B (or both). Use this to prove that $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

Proof. Let x be a limit point of $A \cup B$. Then given any $r > 0$, $B_r(x) \setminus \{x\} \cap (A \cup B) \neq \varnothing$. Then for any such r, $B_r(x)$ intersects A, B, or both. If for all r it intersects both, then clearly x is a limit point of both A and B. If for any r, it only intersects one (say A without loss of generality), then for any $r' < r$, $B_{r'}(x) \setminus \{x\} \subset B_r(x) \setminus \{x\}$, so since $B_{r'}(x) \setminus \{x\}$ intersects $A \cup B$, it must cannot intersect B, and thus it must intersect A. Thus $B_{r'}(x) \setminus \{x\}$ intersects A for all A, and x is a limit point of A. The other direction of the implication is immediate: if x is a limit point of A or B, then it is certainly a limit point of $A \cup B$ since any deleted ball intersecting A or B intersects $A \cup B$.

If $x \in \overline{A \cup B}$, then either $x \in A \cup B$, in which case $x \in \overline{A} \cup \overline{B}$, or x is a limit point of $A \cup B$, in which case x is a limit point of A or B as shown above. On the other hand, if $x \in \overline{A} \cup \overline{B}$, then either x in A or B, in which case $x \in \overline{A \cup B}$, or x is a limit point of A or B, in which case x is a limit point of $A \cup B$ as shown above. $\square$

3. Let $A \subset X$ and recall that $\text{int}(A)$ denotes the interior of A. Show that

$$(\text{int}(A))^c = \overline{(A^c)}.$$

Proof. If $x \in (\text{int}(A))^c$, then $\forall r > 0$, $B_r(x)$ intersects A^c. Thus either $x \in A^c$ or x is a limit point of A^c, so $x \in \overline{(A^c)}$.

Now if $x \in \overline{(A^c)}$, then we have two cases. If $x \in A^c$, then since $\text{int}(A) \subset A$, $x \in (\text{int}(A))^c$. If $x \notin A^c$, then x is a limit point of A^c, so each deleted ball $B_r(x) \setminus \{x\}$ intersects A^c. In particular, there does not exist an r such that $B_r(x) \subset A$, so $x \in (\text{int}(A))^c$. $\square$

4. Show that if d_1 and d_2 are equivalent metrics on the set X, then any set A that is dense in X w.r.t. d_1 is also dense w.r.t. d_2.

Proof. A set $A \subset X$ is dense in X if for every $x \in X$ and every open neighborhood U of x, U intersects A. Since d_1 and d_2 are equivalent, they induce the same open sets on X, so if this condition is met under one metric, it is met under the other. $\square$

5. Let (X, d_X) and (Y, d_Y) be two metric spaces, and let d be the metric on $X \times Y$ defined as in Homework 1, Question 6. Show that if A is dense in X and B is dense in Y, then $A \times B$ is dense in $X \times Y$ w.r.t. d. Use this to show that $\mathbb{Q}^n$ is dense in $\mathbb{R}^n$ for any n (w.r.t. the Euclidean metric).

Proof. Let $(a, b), (c, d) \in X \times Y$. Then for any $r > 0$, $d((a, b), (c, d)) < r \iff d_X(a, c) < r$ and $d_Y(b, d) < r$, so $B_r^{X \times Y}((a, b)) = B_r^X(a) \times B_r^Y(b)$. Thus if A is dense in X and B is dense in Y, then given $U \subset X$ and $V \subset Y$ open sets, U intersects A and V intersects B, so $U \times V$ intersects $A \times B$. Since every ball in $X \times Y$ can be described this way, this shows that $A \times B$ is dense in $X \times Y$.

Induction follows from associativity of cartesian products (e.g., $\mathbb{R}^3 = \mathbb{R}^2 \times \mathbb{R}$), so since $\mathbb{Q}$ is dense in $\mathbb{R}$, we indeed have that $\mathbb{Q}^n$ is dense in $\mathbb{R}^n$ for any n under this "maximum" metric. We have shown that this metric is equivalent to the Euclidean metric on $\mathbb{R}^n$, so we are done. $\square$

6. Let (X, d) be a metric space, and assume that A, B are subsets of X that are both open and dense. Show that $A \cap B$ must be dense.

Note that the conclusion could fail if A, B are not open. For example, $\mathbb{Q}$ and $\mathbb{Q}^c$ are both dense in $\mathbb{R}$ but $\mathbb{Q} \cap \mathbb{Q}^c = \varnothing$.

Proof. Let $U \subset X$ be open and nonempty. Then $U \cap A \neq \varnothing$ since A is dense in X, and $U \cap A$ is open since U and A are both open. We also have that $(U \cap A) \cap B \neq \varnothing$ since B is dense in A, and $(U \cap A) \cap B$ is open since both $U \cap A$ and B are open. Since $(U \cap A) \cap B = U \cap (A \cap B)$ and U was arbitrary, we have that $A \cap B$ is dense in X. $\square$